

ENVS-193DS Final

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Link to Git Hub

[GitHub Repository](#)

Set Up

```
library(tidyverse) # data wrangling and ggplot
library(here)      # file paths
library(janitor)   # data cleaning
library(DHARMA)    # diagnostic checks for models
library(MuMin)    # AIC model selection
library(ggeffects) # model predictions
library(flextable) # making tables
library(rstatix)   # effect size calculations
```

```
nest_data <- read_csv(here("data", "nest_data_final.csv"))
my_data <- read_csv(here("data", "my_data.csv"))
```

Problem 1. Research writing

1a. Transparent statistical methods

Part 1:

In part 1, they used a logistic regression, since the response variable (probability of American Avocet micro habitat use) is binary and distance to water edge is a continuous predictor. ##### Part 2: In part 2, they used a chi-square test of independence, since they are testing the relationship between two categorical variables (bird species and habitat type).

1b. Suggestions for rewriting

Part 1: Distance to water edge was a significant predictor of American Avocet microhabitat use, meaning that where a bird was found was related to how far it was from the water's edge.

Part 2: The three bird species, American Avocet, Forster's Tern, and Black-necked Stilt, were not evenly distributed across habitat types (managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflat), suggesting that each species shows preferences for particular habitats.

1c. Further analysis

Presence of wet bare ground (categorical: present or absent, assessed visually during point count surveys) could influence the probability of American Avocet microhabitat use because, according to Schacter et al. (2023), Avocets fed primarily on wet bare ground in 65% of foraging observations, suggesting they strongly select for this substrate type when choosing where to forage. Sites lacking wet bare ground would therefore be expected to have lower probability of Avocet microhabitat use.

Problem 2. Data analysis

2a. Clean and Wrangle

```
nests_clean <- nest_data |>
  # select only the 3 columns of interest
  select(case_control, height_cm, ant_species) |>
  # filter to only the 3 ant species of interest
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # create column with full scientific names
  mutate(ant_species_full = case_when(
    ant_species == "AB" ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"
  )) |>
  # reorder factor levels: C. sjostedti, C. mimosae, C. nigriceps
  mutate(ant_species_full = fct_relevel(ant_species_full,
    "Crematogaster sjostedti",
    "Crematogaster mimosae",
    "Crematogaster nigriceps"))

# display structure of nests_clean
str(nests_clean)
```

tibble [270 x 4] (S3: tbl_df/tbl/data.frame)

```
$ case_control      : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
$ height_cm        : num [1:270] 392 310 513 154 324 ...
$ ant_species       : chr [1:270] "RRB" "RRB" "RRB" "RRB" ...
$ ant_species_full : Factor w/ 3 levels "Crematogaster sjostedti",...: 2 2 2 2 1 2 3 2 1 3 ..
```

```
# display 10 random rows
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
```

	case_control	height_cm	ant_species	ant_species_full
	<dbl>	<dbl>	<chr>	<fct>
1	0	168	RRB	Crematogaster mimosae
2	0	118.	RRB	Crematogaster mimosae
3	1	389	RRB	Crematogaster mimosae
4	0	123	RRB	Crematogaster mimosae
5	0	152.	BBR	Crematogaster nigriceps
6	0	156	AB	Crematogaster sjostedti
7	0	513	RRB	Crematogaster mimosae
8	1	212.	RRB	Crematogaster mimosae
9	0	199	RRB	Crematogaster mimosae
10	0	178	RRB	Crematogaster mimosae

2b. Response variable

In this dataset, a 1 indicates that the tree was a “used” tree, meaning a bird nest of any species was found in that tree. A 0 indicates that the tree was an “available” tree, meaning it was one of the four nearest neighbors to a nest tree but did not itself contain a nest.

2c. Purpose of study

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence would increase because taller trees may offer greater structural complexity and concealment from predators, making them more suitable nest sites; the authors found that for each 1 m increase in height, the odds that birds nested in a tree increased by 20%. This information was found in the Methods section (paragraph 2) and the Results section (paragraph 1).

Acacia ant species is a categorical variable with three levels: Crematogaster sjostedti, Crematogaster mimosae, and Crematogaster nigriceps. As ant species identity shifts from less aggressive defenders (C. sjostedti) to more aggressive defenders (C. mimosae and C. nigriceps), the probability of nest occurrence would increase because more aggressive ant species vigorously

defend host trees against mammalian herbivores and predators, which may also protect bird nests from disturbance and predation. This information was found in the Introduction (paragraphs 2–3) and the Discussion (paragraphs 1–2).

2d. Table of Models

```
model_table <- data.frame(
  "Model Number" = c("Model 1", "Model 2"),
  "Response Variable" = c("Probability of nest occurrence",
                           "Probability of nest occurrence"),
  "Predictor Variables" = c("Tree height (cm), ant species",
                             "Tree height (cm), ant species"),
  "Interaction Term" = c("No", "Yes (height × ant species)")
)

flextable(model_table) |>
  set_header_labels(
    Model.Number = "Model Number",
    Response.Variable = "Response Variable",
    Predictor.Variables = "Predictor Variables",
    Interaction.Term = "Interaction Term"
  ) |>
  align(align = "center", part = "all") |>
  bold(part = "header") |>
  set_table_properties(width = 1, layout = "autofit") |>
  fontsize(size = 9, part = "all")
```

Model Number	Response Variable	Predictor Variables	Interaction Term
Model 1	Probability of nest occurrence	Tree height (cm), ant species	No
Model 2	Probability of nest occurrence	Tree height (cm), ant species	Yes (height × ant species)

2e. Run models

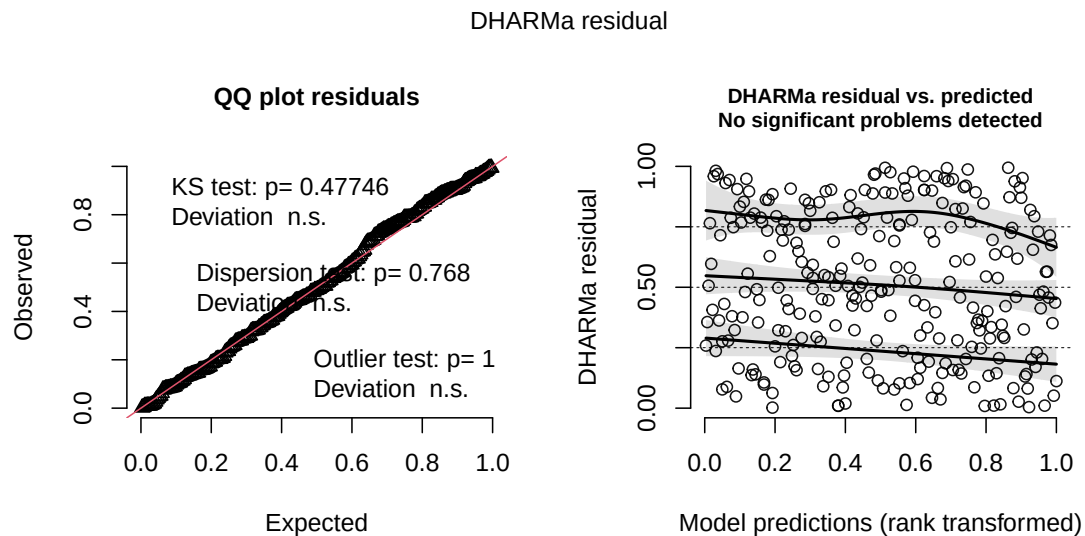
```
# model 1: tree height + ant species, no interaction
model1 <- glm(case_control ~ height_cm + ant_species_full,
              data = nests_clean,
              family = binomial)
```

```
# model 2: tree height * ant species, with interaction
model2 <- glm(case_control ~ height_cm * ant_species_full,
              data = nests_clean,
              family = binomial)
```

2f. Select the best model

The best model as determined by Akaike's Information Criterion (AIC) was the model with an interaction term between tree height and acacia ant species, predicting the probability of bird nest occurrence ($AICc = 238.12$). This model performed better than the additive model with no interaction term ($AICc = 258.89$, $\Delta AICc = 20.77$), indicating that the effect of tree height on the probability of nest occurrence differs depending on which acacia ant species occupies the tree. A $\Delta AICc$ greater than 2 indicates meaningful improvement in model fit, suggesting the interaction term is an important component of the model. ### 2g. Check diagnostics

```
# simulate residuals from best model (model 2, with interaction)
simulation_output <- simulateResiduals(fittedModel = model2,
                                       plot = TRUE)
```

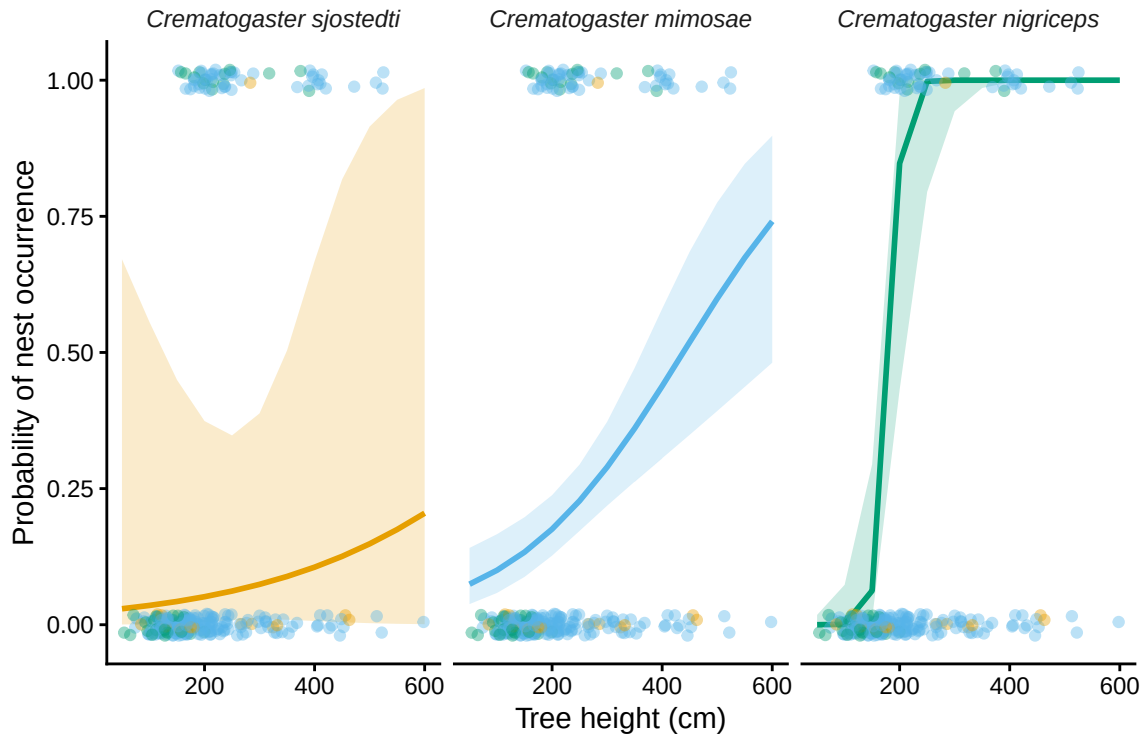


The QQ plot shows simulated residuals closely following the expected distribution, and the residual vs. predicted plot shows no significant problems detected (KS test: $p = 0.48$, dispersion test: $p = 0.77$, outlier test: $p = 1$). The model assumptions are met and the model is appropriate for inference.

2h. Visualize the model predictions

```
# generate predictions from model 2 for height across all 3 ant species
predictions <- ggpredict(model2,
                          terms = c("height_cm", "ant_species_full"))

# plot predictions with underlying data
ggplot(predictions, aes(x = x, y = predicted, color = group, fill = group)) +
  # 95% confidence interval ribbon
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
             alpha = 0.2,
             color = NA) +
  # model prediction line
  geom_line(linewidth = 1) +
  # underlying data points, jittered slightly so 0s and 1s are visible
  geom_jitter(data = nests_clean,
             aes(x = height_cm,
                 y = case_control,
                 color = ant_species_full),
             height = 0.02,
             alpha = 0.4,
             inherit.aes = FALSE) +
  # separate panel per ant species
  facet_wrap(~group) +
  # custom colorblind-friendly colors
  scale_color_manual(values = c("Crematogaster sjostedti" = "#E69F00",
                                "Crematogaster mimosae"   = "#56B4E9",
                                "Crematogaster nigriceps"  = "#009E73")) +
  scale_fill_manual(values = c("Crematogaster sjostedti" = "#E69F00",
                                "Crematogaster mimosae"   = "#56B4E9",
                                "Crematogaster nigriceps"  = "#009E73")) +
  # axis labels
  labs(x = "Tree height (cm)",
       y = "Probability of nest occurrence") +
  # clean theme, no gridlines, no legend
  theme_classic() +
  theme(legend.position = "none",
        strip.text = element_text(face = "italic"),
        strip.background = element_blank())
```



```
# calculate predicted probability of nest occurrence at 600 cm for each ant species
predict_600 <- ggpredict(model2,
  terms = c("height_cm [600]",
    "ant_species_full"))

# display output
predict_600
```

```
# Predicted probabilities of case_control
```

```
ant_species_full: Crematogaster sjostedti
```

height_cm	Predicted	95% CI
600	0.20	0.00, 0.99

```
ant_species_full: Crematogaster mimosae
```

height_cm	Predicted	95% CI
-----------	-----------	--------

600		0.74		0.48, 0.90
-----	--	------	--	------------

ant_species_full: *Crematogaster nigriceps*

height_cm		Predicted		95% CI

600		1.00		1.00, 1.00

2i. Write Caption for figure

Figure 1. Predicted probability of bird nest occurrence as a function of tree height for three acacia ant species in a Kenyan savanna. Lines show model predicted probability of nest occurrence across a gradient of tree heights (cm), with shaded ribbons representing 95% confidence intervals. Points show observed presence (1) or absence (0) of bird nests, jittered slightly for visibility. Each panel represents a different acacia ant species occupying the tree (*Crematogaster sjostedti*, *Crematogaster mimosae*, *Crematogaster nigriceps*). Data from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

2j. Calculate model predictions

```
# calculate predicted probability of nest occurrence at 600 cm for each ant species
predict_600 <- ggpredict(model2,
                          terms = c("height_cm [600]",
                                     "ant_species_full"))

# display output
predict_600
```

Predicted probabilities of case_control

ant_species_full: *Crematogaster sjostedti*

height_cm		Predicted		95% CI

600		0.20		0.00, 0.99

ant_species_full: *Crematogaster mimosae*

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

ant_species_full: *Crematogaster nigriceps*

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

2k. Interpret your results

As shown in Figure 1, the probability of bird nest occurrence increased with tree height for trees occupied by *Crematogaster mimosae* and *Crematogaster nigriceps*, while trees occupied by *Crematogaster sjostedti* showed little increase in nest probability with height. At 600 cm, the tallest tree in the dataset, the predicted probability of nest occurrence was 0.20 for *C. sjostedti*, 0.74 for *C. mimosae*, and 1.00 for *C. nigriceps* trees (Figure 1). The substantially higher probability of nest occurrence in trees occupied by *C. mimosae* and *C. nigriceps* is likely explained by the fact that these species aggressively defend their host trees against mammalian herbivores and predators, which may also protect bird nests from disturbance and predation (Lujan et al. 2023). Additionally, *C. nigriceps* prunes apical buds to create denser canopies, which likely provides birds with additional concealment from nest predators such as snakes and raptors (Lujan et al. 2023). In contrast, *C. sjostedti* provides only minimal protection against herbivory, making those trees far less attractive nest sites regardless of tree height (Lujan et al. 2023).

Problem 3. Affective and exploratory visualizations

3a. Comparing visualizations

How are the visualizations different? The exploratory visualizations from Homework 2 used standard statistical plot types, a jitter plot with means and error bars comparing focus level between work and non-work days and a scatterplot of water intake versus focus level, while the affective visualization from Homework 3 represented each day as a hand-drawn drinking glass where fill level encoded water intake and water color (green to blue) encoded focus level, inspired by the Dear Data project. The exploratory plots prioritize precise visual comparison of values across groups, while the affective visualization prioritizes an embodied, personal feeling of each day's energy and hydration.

What similarities do you see? All visualizations represent the same two core variables — daily water intake and focus level — and all show individual observations rather than aggregated summaries alone. Both the exploratory and affective visualizations also incorporate work day as an additional variable, encoded through color grouping in the exploratory plots and through sun or moon symbols in the affective visualization.

What patterns do you see? In the Homework 2 exploratory plots, focus level appeared slightly lower on work days compared to non-work days, and higher water intake appeared associated with higher focus level. The effective visualization from Homework 3 showed the same general pattern — glasses with more water fill tended to have cooler, bluer colors indicating higher focus. The patterns are consistent across visualizations because all draw from the same underlying data set, though the effective visualization makes the day-to-day variation feel more personal and tangible than the summary statistics in the exploratory plots.

What feedback did you get in week 9? In week 9 workshop, feedback suggested making the legend clearer so viewers could decode what the fill level and water color represented without prior context. I tried to implement this by including a hand-drawn legend on the back of the postcard indicating what fill levels and colors corresponded to specific water intake and focus values, though the legend remains somewhat informal given the hand-drawn medium of the piece.

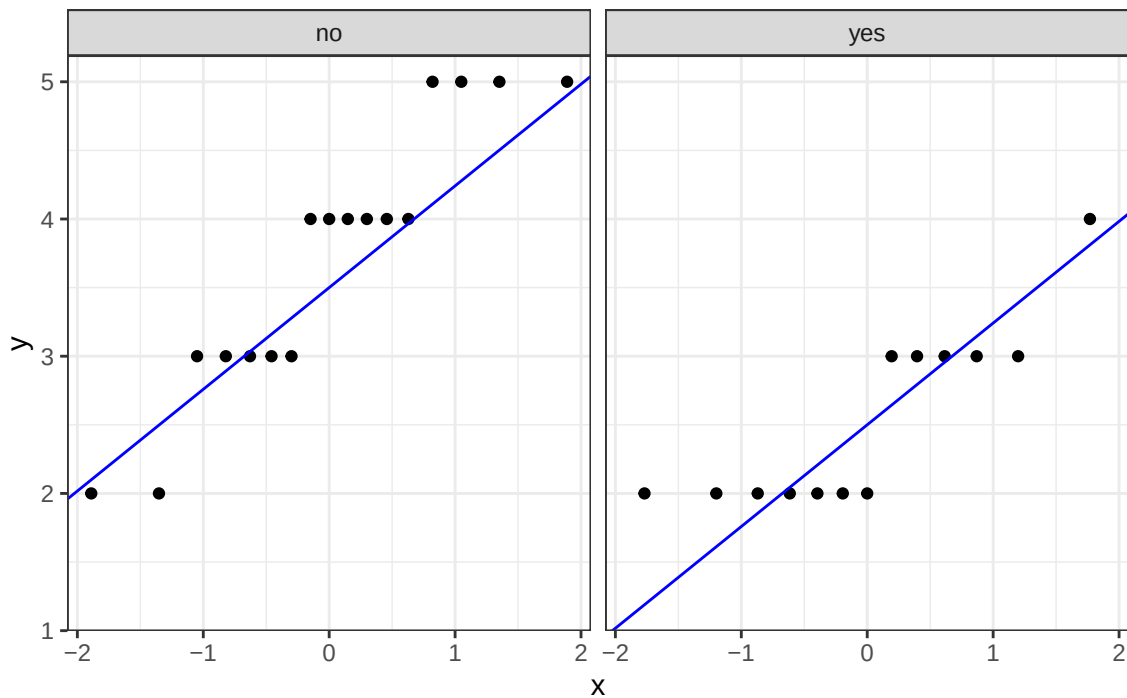
3b. Designing an analysis

I propose a Welch's two-sample t-test comparing mean focus level between work days and non-work days. This analysis is appropriate because my research question asks whether focus level differs depending on whether the day is a work day; my response variable (focus level, 1–5 scale) is continuous, and my predictor variable (work day: yes/no) is categorical with two levels. A t-test is the appropriate method for comparing means between two independent groups when the response variable is continuous and approximately normally distributed, which I will verify with QQ plots before running the test.

3c. Checking assumptions

```
# check normality of focus level for each work day group using QQ plots
ggplot(data = my_data,
       aes(sample = focus_level)) +
  geom_qq() +
  geom_qq_line(color = "blue") +
  facet_wrap(~work_day) +
  labs(title = "QQ plots: Focus level by work day") +
  theme_bw()
```

QQ plots: Focus level by work day



Running the analysis

```
# run Welch's two-sample t-test comparing focus level between work and non-work days
t_test_result <- t.test(focus_level ~ work_day,
                        data = my_data,
                        var.equal = FALSE)

# display result
t_test_result
```

Welch Two Sample t-test

```
data: focus_level by work_day
t = 3.878, df = 27.613, p-value = 0.0005929
alternative hypothesis: true difference in means between group no and group yes is not equal
95 percent confidence interval:
 0.5503826 1.7844591
```

sample estimates:

mean in group no	mean in group yes
3.705882	2.538462

Effect size

```
# calculate Cohen's d effect size to quantify magnitude of difference
cohens_d(focus_level ~ work_day,
  data = my_data)
```

A tibble: 1 x 7

.y.	group1	group2	effsize	n1	n2	magnitude	
* <chr>	<chr>	<chr>	<dbl>	<int>	<int>	<ord>	
1	focus_level	no	yes	1.39	17	13	large

3d. Visualization

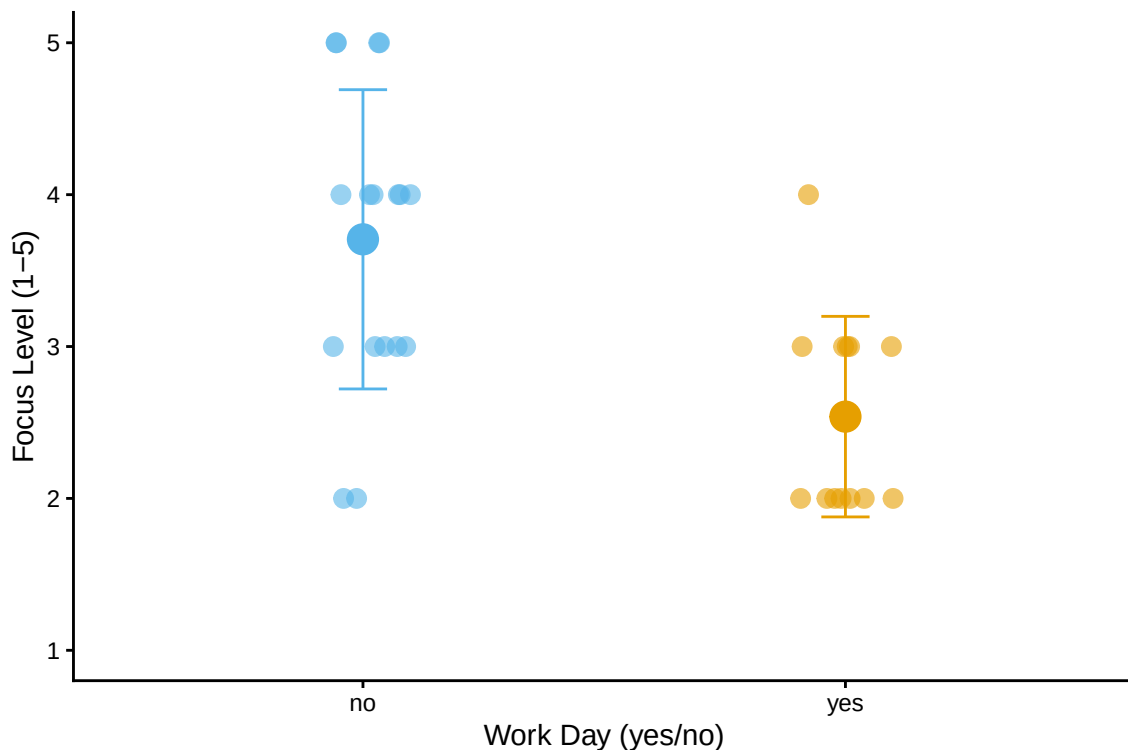
```
# summarize mean and SD by work day for plotting
my_data_summary <- my_data |>
  group_by(work_day) |>
  summarize(mean_focus = mean(focus_level),
    sd_focus = sd(focus_level))

# plot focus level by work day with raw data, means, and SD error bars
ggplot() +
  geom_jitter(data = my_data,
    aes(x = work_day,
      y = focus_level,
      color = work_day),
    width = 0.1, height = 0,
    size = 3, alpha = 0.6) +
  geom_point(data = my_data_summary,
    aes(x = work_day,
      y = mean_focus,
      color = work_day),
    size = 5) +
  geom_errorbar(data = my_data_summary,
    aes(x = work_day,
      ymin = mean_focus - sd_focus,
```

```

      ymax = mean_focus + sd_focus,
      color = work_day),
      width = 0.1) +
scale_color_manual(values = c("no" = "#56B4E9",
                              "yes" = "#E69F00")) +
scale_y_continuous(limits = c(1, 5),
                  breaks = c(1, 2, 3, 4, 5)) +
labs(x = "Work Day (yes/no)",
     y = "Focus Level (1-5)") +
theme_classic() +
theme(legend.position = "none")

```



3e. Write a caption

Figure 2. Daily focus level (1-5 scale) on work days versus non-work days across 30 days of personal observation. Small points show individual daily focus level observations, jittered horizontally to reduce overlap. Large points show the group mean and error bars represent plus or minus 1 standard deviation. Blue points represent non-work days and orange points represent work days.

3f. Write about your results

Non-work days had a higher mean focus level (mean = 3.71, n = 17) compared to work days (mean = 2.54, n = 13), a difference of about 1.17 points on the 1-5 scale. This difference was statistically significant and large in magnitude (Welch's t-test: $t(27.61) = 3.88$, $p < 0.001$, Cohen's $d = 1.39$), suggesting that whether or not a day is a work day has a strong association with how focused I feel. This pattern may reflect the increased demands and reduced autonomy over daily schedule that come with work days, which could make it harder to sustain attention and mental clarity throughout the day.